Beyond the Manual: Examining the Structure of the BASC-3 in Neurodevelopmental and Standardization Samples

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BACKGROUND

- The Behavioral Assessment System for Children (BASC) is commonly for behavioral and emotional concerns in youth
- Often used as **part of assessment and diagnosis for kids with neurodevelopmental disorders (NDDs)** [3]
- The factor structure has not been independently validated in NDD populations in the BASC-2 or -3 [2]
  - ➢ Known-groups validity and classification accuracy has been evaluated, but results are mixed [1]
  - ➢ Does not guarantee that the scores can be interpreted the same in kids with NDD

OBJECTIVES

1) Evaluate the fit of the hypothesized structures published in the BASC-3 manual and explore new structures in four diverse, independent samples.
2) Compare exploratory structures and model fit between samples.

METHOD

- Four samples of the BASC-3 PRS-C (Table 1):
  - ❖ **Referred NDD**: Children referred to the neurodevelopmental clinic in a large hospital in Central Ohio (n = 764; M age = 8.73 years; 64.7% male; 77.8% White)
  - ❖ **General Norm**: Standardization sample data from the BASC-3 Manual [4] (n = 600; M age = 8.04 years; 50.0% male; 52.3% White, 13.5% African American, 24.3% Hispanic)
  - ❖ **Australian**: Referred general mental health data from Australian university clinic (n = 428; M age = 8.64 years; 59.3% male) [5]
  - ❖ **School**: School-based assessment data from large urban Southwestern school district (n = 152; M age = 8.45 years; 67.8% male; 32.2% White, 21.7% African American, 34.2% Hispanic) [6]
- **Confirmatory factor analyses (CFA)** were conducted for the scoring structure and CSA-derived three-factor structure from the manual [4]
- Models for each were built with **exploratory structural equation modeling (ESEM)**, using parallel analysis to determine the appropriate number of factors (Figure 2)

Parent-report BASC-3 factor structure showed **limited replication** in four independent samples for children, suggesting that further evaluation of its **diagnostic utility** may be warranted.

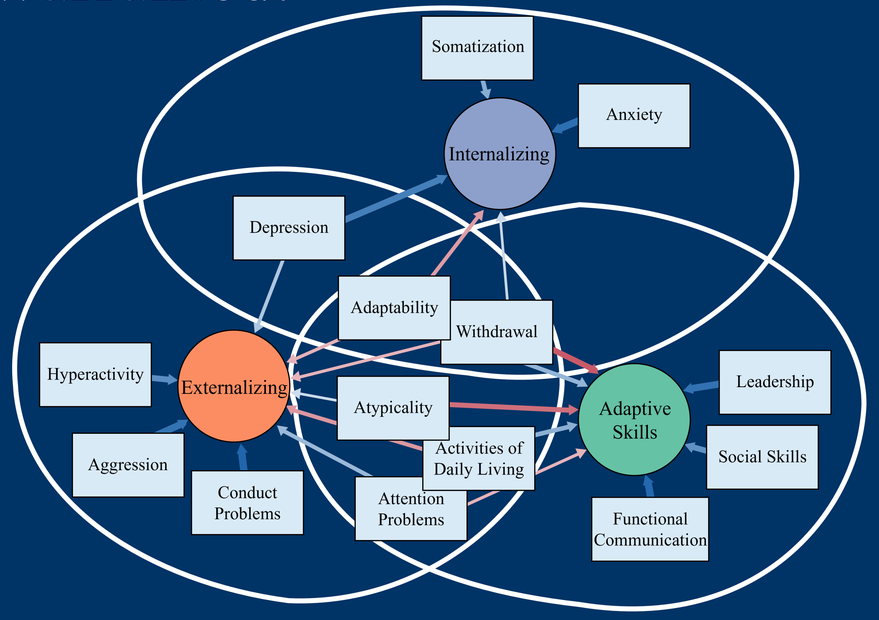


Figure 1. **Representative ESEM Factor Structure Venn Diagram From All Samples.**

[References] and Supplementary Information →

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RESULTS

	CFI	TLI	RMSEA	Fit Evaluation
CFA				
Scoring Structure	.731 - .872	.641 - .829	.138 - .178	👎
Manual CSA Structure	.714 - .856	.634 - .815	.144 - .180	👎
ESEM	.848 - .932	.733 - .881	.115 - .154	👎

Table 1. **Fit Statistics Summary.** Three fit indices were calculated for each model in each sample: Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), and Root Mean Square Error of Approximation (RMSEA). Ranges for these indices in all the samples are shown in the boxes, with full fit tables in Supplementary Information. **None of the fit indices reached dynamic or conventional fit cut-offs** [7-9] (Supp. Information).

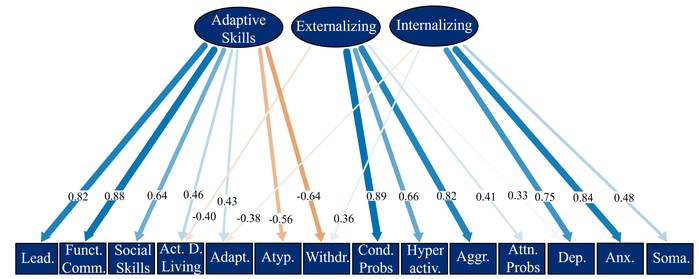


Figure 2. **Referred NDD ESEM Diagram.**

CONCLUSIONS

- Scoring and CSA structures published in the BASC-3 manual shows **poor fit** in all samples (Table 1)
- **A three-factor solution shows slightly better fit**, but the subscales load inconsistently across samples
- Practitioners should refrain from making decisions solely based on composite scores

Recommendations: Focus on subscale scores and favor specialized measures of adaptive skills

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